

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030414

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Query 1: Create the database

```
CREATE DATABASE Bank_ACID_DB1;  
USE Bank_ACID_DB1;
```

Output: -

#	Time	Action	Message	Duration / Fetch
✓ 1	19:06:12	CREATE DATABASE Bank_ACID_DB1	1 row(s) affected	0.031 sec
✓ 2	19:06:14	USE Bank_ACID_DB1	0 row(s) affected	0.000 sec

Query 2: -CREATE CUSTOMER TABLE

```
CREATE TABLE Customer (  
    Customer_ID INT PRIMARY KEY,  
    Customer_Name VARCHAR(100) NOT NULL,  
    Phone VARCHAR(15),  
    City VARCHAR(50)  
);
```

Output: -

✓ 3	19:07:15	CREATE TABLE Customer (Customer_ID INT PR...	0 row(s) affected	0.079 sec
-----	----------	---	-------------------	-----------

Query 3: -CREATE ACCOUNT TABLE

```
CREATE TABLE Account (  
    Account_No INT PRIMARY KEY,  
    Customer_ID INT,  
    Account_Type VARCHAR(20),  
    Balance DECIMAL(12,2),  
    Branch VARCHAR(50),  
  
    FOREIGN KEY (Customer_ID)  
    REFERENCES Customer(Customer_ID)  
);
```

Output: -

✓ 4 19:08:34 CREATE TABLE Account (Account_No INT PRI... 0 row(s) affected 0.094 sec

Query 4: -CREATE TRANSACTION TABLE

```
CREATE TABLE Bank_Transaction (  
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,  
    Account_No INT,  
    Transaction_Type VARCHAR(20),  
    Amount DECIMAL(12,2),  
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,  
  
    FOREIGN KEY (Account_No)  
    REFERENCES Account(Account_No)  
);
```

Output: -

✓ 5 19:09:46 CREATE TABLE Bank_Transaction (Transaction... 0 row(s) affected

Query 5: -INSERT CUSTOMER DATA

```
1 • INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City)  
2 VALUES  
3 (101, 'YISHAANI', '1234535235', 'Hyderabad'),  
4 (102, 'RAHUL', '9876543210', 'Mumbai'),  
5 (103, 'PRIYA', '9123456789', 'Delhi'),  
6 (104, 'ARJUN', '9988776655', 'Bangalore'),  
7 (105, 'ANANYA', '8765432109', 'Chennai'),  
8 (106, 'ROHAN', '7894561230', 'Pune'),  
9 (107, 'SAI', '9345678120', 'Hyderabad'),  
10 (108, 'NEHA', '8899776655', 'Kolkata'),  
11 (109, 'VIKRAM', '7654321098', 'Jaipur'),  
12 (110, 'KAVYA', '9012345678', 'Vijayawada'),  
13 (111, 'ADITYA', '8123456789', 'Delhi'),  
14 (112, 'SNEHA', '9988123456', 'Mumbai'),  
15 (113, 'KARTHIK', '9345126789', 'Chennai'),  
16 (114, 'MEGHA', '8765123490', 'Pune'),
```

Output: -

✓ 6 19:12:51 INSERT INTO Customer (Customer_ID, Customer_N... 5 row(s) affected Records: 5 Duplicates: 0 Warning... 0.031 sec

Query 6: -INSERT ACCOUNT DATA

```

INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),

(10002, 102, 'Savings', 75000, 'Vijayawada'),

(10003, 103, 'Current', 120000, 'Bangalore'),

(10004, 104, 'Savings', 45000, 'Chennai'),

(10005, 105, 'Current', 90000, 'Hyderabad');

```

Output: -

7 19:14:08 INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.032 sec

Query 7: -INSERT BANK TRANSACTION DATA

```

INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),

(10002, 'DEPOSIT', 15000),

(10003, 'WITHDRAW', 20000),

(10004, 'DEPOSIT', 5000),

(10005, 'WITHDRAW', 10000);

```

Output: -

8 19:15:22 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.032 sec

Query 8: -DISPLAY DATA

```

SELECT * FROM Customer;

```

Output: -

	103	Anirudh Reddy	9876543212	Bangalore
	104	Akshara Reddy	9876543213	Chennai
	105	Ananya Varshita Reddy	9876543214	Hyderabad
*	NULL	NULL	NULL	NULL

Customer 1 x

Apply

Output



Action Output

	#	Time	Action	Message
✓	4	19:08:34	CREATE TABLE Account (Account_No INT P...	0 row(s) affected
✓	5	19:09:46	CREATE TABLE Bank_Transaction (Transacti...	0 row(s) affected
✓	6	19:12:51	INSERT INTO Customer (Customer_ID, Customer...	5 row(s) affected Re
✓	7	19:14:08	INSERT INTO Account (Account_No, Customer_...	5 row(s) affected Re
✓	8	19:15:22	INSERT INTO Bank_Transaction (Account_No, ...	5 row(s) affected Re
✓	9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned

SELECT * FROM Account;

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
•	NULL	NULL	NULL	NULL	NULL

Account 2 ×

Apply

Output



Action Output

	#	Time	Action	Message
✓	5	19:09:46	CREATE TABLE Bank_Transaction (Transacti...	0 row(s) affected
✓	6	19:12:51	INSERT INTO Customer (Customer_ID, Customer...	5 row(s) affected R
✓	7	19:14:08	INSERT INTO Account (Account_No, Customer_...	5 row(s) affected R
✓	8	19:15:22	INSERT INTO Bank_Transaction (Account_No, ...	5 row(s) affected R
✓	9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓	10	19:17:00	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned

SELECT * FROM Bank_Transaction;

	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
▶	1	10001	DEPOSIT	10000.00	2026-08-15 19:15:22
	2	10002	DEPOSIT	15000.00	2026-08-15 19:15:22
	3	10003	WITHDRAW	20000.00	2026-08-15 19:15:22
	4	10004	DEPOSIT	5000.00	2026-08-15 19:15:22
	5	10005	WITHDRAW	10000.00	2026-08-15 19:15:22
•	NULL	NULL	NULL	NULL	NULL

Bank_Transaction 3 ×

Output

📄 Action Output ▼

	#	Time	Action
✓	6	19:12:51	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'Asw
✓	7	19:14:08	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VA
✓	8	19:15:22	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10
✓	9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000
✓	10	19:17:00	SELECT * FROM Account LIMIT 0, 1000
✓	11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000

Query9: -START A TRANSACTION

70	•	START TRANSACTION;
71		
72		
73		
74		

Output			
Action Output			
#	Time	Action	Message
✓ 7	19:14:08	INSERT INTO Account (Account_No, Customer_...	5 row(s) affected Reco
✓ 8	19:15:22	INSERT INTO Bank_Transaction (Account_No, ...	5 row(s) affected Reco
✓ 9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 10	19:17:00	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 12	19:19:30	START TRANSACTION	0 row(s) affected

Query 10 : - DEPOSIT MONEY

71	•	UPDATE Account
72		SET Balance = Balance + 5000
73		WHERE Account_No = 10001;
74		

Output			
Action Output			
#	Time	Action	Message
✓ 8	19:15:22	INSERT INTO Bank_Transaction (Account_No, ...	5 row(s) affected Records: 5 Duplicates: 0 Wami
✓ 9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 10	19:17:00	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 12	19:19:30	START TRANSACTION	0 row(s) affected
✓ 13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 .

Query 11: -CHECK BALANCE

74 •

SELECT *

75

FROM Account

76

WHERE Account_No = 10001;

77

Result Grid

Filter Rows:

Edit:

Export/Import

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Account 4

Apply

Revert

Context H

Output

Action Output

#	Time	Action	Message
✓ 9	19:16:44	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
✓ 10	19:17:00	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓ 11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 12	19:19:30	START TRANSACTION	0 row(s) affected
✓ 13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Change
✓ 14	19:21:27	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned

Query 12: -COMMIT TRANSACTION

77 • COMMIT;

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Output



Action Output

	#	Time	Action	Message
✓	10	19:17:00	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
✓	11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓	12	19:19:30	START TRANSACTION	0 row(s) affected
✓	13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows
✓	14	19:21:27	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
✓	15	19:22:45	COMMIT	0 row(s) affected

Query 13: -CHECK BALANCE AFTER COMMIT

78 • `SELECT *`
 79 `FROM Account`
 80 `WHERE Account_No = 10001;`
 81

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
•	NULL	NULL	NULL	NULL	NULL

Account 5 x Apply Revert Context Help

Output

Action Output

#	Time	Action	Message
✓ 11	19:17:16	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
✓ 12	19:19:30	START TRANSACTION	0 row(s) affected
✓ 13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...
✓ 14	19:21:27	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
✓ 15	19:22:45	COMMIT	0 row(s) affected
✓ 16	19:23:29	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned

Query 14: -ROLLBACK

START TRANSACTION

81 • **START TRANSACTION;**

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[Context Help](#) [Snippets](#)

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 12	19:19:30	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
✓ 14	19:21:27	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 15	19:22:45	COMMIT	0 row(s) affected	0.031 sec
✓ 16	19:23:29	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 17	19:26:05	START TRANSACTION	0 row(s) affected	0.000 sec

Query 15: -WITHDRAW MONEY

82 • **UPDATE Account**

83 **SET Balance = Balance - 10000**

84 **WHERE Account_No = 10001;**

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[Context Help](#) [Snippets](#)

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 13	19:20:37	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
✓ 14	19:21:27	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 15	19:22:45	COMMIT	0 row(s) affected	0.031 sec
✓ 16	19:23:29	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 17	19:26:05	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec

Query 16: -CHECK BALANCE

85 • `SELECT *`
 86 `FROM Account`
 87 `WHERE Account_No = 10001;`

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	45000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Account 6 x Apply Revert Context Help

Output

Action Output

#	Time	Action	Message
✓ 14	19:21:27	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
✓ 15	19:22:45	COMMIT	0 row(s) affected
✓ 16	19:23:29	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
✓ 17	19:26:05	START TRANSACTION	0 row(s) affected
✓ 18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...
✓ 19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned

Query 17: -ROLLBACK & CHECK BALANCE AFTER ROLLBACK

88 • ROLLBACK;
 89 • SELECT *
 90 FROM Account
 91 WHERE Account_No = 10001;

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	55000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Form Editor
Field Types

Account 7 x Apply Revert Context Help

Output

Action Output

#	Time	Action	Message
✓ 16	19:23:29	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
✓ 17	19:26:05	START TRANSACTION	0 row(s) affected
✓ 18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...
✓ 19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned
✓ 20	19:29:31	ROLLBACK	0 row(s) affected
✓ 21	19:29:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned

Query 18: -SAVEPOINT

92 • **START TRANSACTION;**

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[Context Help](#) [Snippets](#)

Output

 Action Output

#	Time	Action	Message	Duration / Fetch
✓ 17	19:26:05	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
✓ 19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 20	19:29:31	ROLLBACK	0 row(s) affected	0.000 sec
✓ 21	19:29:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 22	19:31:11	START TRANSACTION	0 row(s) affected	0.000 sec

Query 19: -FIRST UPDATE

93 • **UPDATE Account**

94 **SET Balance = Balance + 5000**

95 **WHERE Account_No = 10001;**

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101

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[Context Help](#) [Snippets](#)

Output

 Action Output

#	Time	Action	Message	Duration / Fetch
✓ 18	19:27:37	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
✓ 19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 20	19:29:31	ROLLBACK	0 row(s) affected	0.000 sec
✓ 21	19:29:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 22	19:31:11	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 23	19:32:04	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec

Query 20: -CREATE SAVEPOINT

```
96 • SAVEPOINT Deposit1;
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99
100
101
102
103
104
105
106
```

Context Help Snippets

Output :

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 19	19:28:40	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 20	19:29:31	ROLLBACK	0 row(s) affected	0.000 sec
✓ 21	19:29:56	SELECT * FROM Account WHERE Account_No ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 22	19:31:11	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 23	19:32:04	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 24	19:33:03	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec

Query 21: -SECOND UPDATE

```
UPDATE Account
SET Balance = Balance - 3000
WHERE Account_No = 10002;
```

✓ 26	19:34:14	UPDATE Account SET Balance = Balance - 300...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
------	----------	---	--	-----------

Query 22: -CREATE SECOND SAVEPOINT

```
SAVEPOINT Withdrawal1;
```

✓ 25	19:33:39	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
------	----------	-----------------------	-------------------	-----------

Query 23: -ROLLBACK TO SAVEPOINT

101 • **ROLLBACK TO SAVEPOINT Withdrawal1;**

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[Context Help](#) [Snippets](#)

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 22	19:31:11	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 23	19:32:04	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 24	19:33:03	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
✓ 25	19:33:39	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
✓ 26	19:34:14	UPDATE Account SET Balance = Balance - 300...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 27	19:36:33	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec

Query 24: -COMMIT

102 • **COMMIT;**

103

104

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106

[Context Help](#) [Snippets](#)

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 23	19:32:04	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 24	19:33:03	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
✓ 25	19:33:39	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
✓ 26	19:34:14	UPDATE Account SET Balance = Balance - 300...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 27	19:36:33	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
✓ 28	19:37:23	COMMIT	0 row(s) affected	0.016 sec

Query 25: -CHECK ACCOUNTS

103 • SELECT *

104 FROM Account;

105

Result Grid

Filter Rows:

Edit:

Export/Im

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	60000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
•	NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Account 8 x

Apply

Revert

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 24	19:33:03	SAVEPOINT Deposit1	0 row(s) affected	0.000 sec
✓ 25	19:33:39	SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
✓ 26	19:34:14	UPDATE Account SET Balance = Balance - 300...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 27	19:36:33	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected	0.000 sec
✓ 28	19:37:23	COMMIT	0 row(s) affected	0.016 sec
✓ 29	19:37:54	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

toggle automatic help.